SERVICE PROPOSAL

NPIC QC & Operations System

Phase 2 — Production & Packaging

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1. Overview

This proposal outlines the scope of work for **Phase 2** of the NPIC digital operations platform. Building on the QC inspection system established in Phase 1 (Modules A–I), Phase 2 extends the platform to cover the full production floor, dry room, and packaging department — replacing the current Excel spreadsheets and paper-based forms with a unified, role-controlled digital system.

Phase 2 introduces ten new functional modules (A–J) that together provide end-to-end traceability from raw material (一级单) assignment through production, drying, packaging, finished goods inventory, employee attendance and performance, and document management.

2. Current State & Challenges

Area	Current Method	Key Challenges
Production Tracking	Excel daily report (2026 Daily Report Forming Production.xlsx)	Manual entry, error-prone, no real-time visibility, productivity calculated by hand
Dry Room	Paper-based cart logs	No systematic cart-chain visibility; half-cart continuations are hard to track
Packaging — Material Pick	Handwritten Picking Sheet	No digital record; lot number reconciliation is manual
Packaging — Shift Record	Handwritten Traveler Form (Packing)	Multiple paper copies; no automatic totals; supervisor signature on paper only
Packaging — Pallet Record	Handwritten Pallet Traveler Form	QC results (moisture, water activity, metal detection) stored on paper only
Testing & Release	Paper-based	No traceability; pass/fail decisions and redry routing are manual
Finished Goods / Warehouse	Manual	FIFO not enforced digitally; lot tracking is inconsistent
Sage 100 Data Entry	All paper records manually re-keyed into Sage 100 by staff	Time-consuming, error-prone double-entry; delays between production and system records

3. System Architecture

Phase 2 is built on the same technical stack as Phase 1 and integrates directly with the existing NPIC platform. New modules share the same user authentication, barcode scanning, and data export infrastructure.

Role & Access Design

Role	Modules Accessible	MFA Required?
Employee	Module B — Station Mode (sign-in/out, cart entry)	No
Supervisor	Modules A, B, C, D, E, F — full entry and review	No (standard operations)
Manager	All modules + restricted actions (work order close, release authorization)	Yes — Microsoft Authenticator
Admin	Full system access including configuration and permission management	Yes — Microsoft Authenticator

Key Design Principles

- **Employee Mode** — Lightweight interface for the production floor. Badge scan or employee ID sign-in. No MFA required for daily entries.
- **Supervisor Mode** — Full data entry and batch logging, matching the existing Excel form layout.
- **Privileged Actions** — Work order closing, release authorization, and other high-impact operations require Microsoft Authenticator verification (inherited from Phase 1, Module I).
- **SKU-Configurable Fields** — Required fields (weight vs. quantity, required work order fields) are pre-configured per product SKU and applied automatically at entry time.
- **Half-Cart Linking** — A structured mechanism to link physical cart continuations across sessions, ensuring unbroken traceability.

4. Module Specifications

Module A Production — Supervisor Data Entry

Replaces the existing Excel daily report. Supervisors enter batch records covering all workers and carts for a given shift. This mode serves teams that do not have individual barcode scanning stations.

Data Captured

- Employee names and IDs
- Machine assignments
- Work order numbers (工单号)
- Primary material codes — 一级单 (raw material identifiers)

- Cart numbers, and quantity or weight per cart
- Shift hours and total labor time

Outputs

- Auto-calculated per-employee productivity metrics
- Shift summary reports, exportable by date range or product (leveraging Phase 1 export framework)

Module B Production — Employee Station Mode

A streamlined, employee-facing interface for real-time production logging at the machine or line level. Designed for use on shared tablets or terminals on the production floor. Microsoft Authenticator is not required for standard daily entries.

Sign-In / Sign-Out

- Employees sign in before the shift starts by entering their employee ID or scanning their badge. Multiple employees can sign in simultaneously on the same machine.
- A digital signature is required at sign-out.
- The shift is closed when all assigned employees on that line have signed out.

Cart Entry

- On the first cart, the employee inputs the primary work order number(s). Subsequent carts auto-populate from the last entry, but a **confirmation button must be pressed** before proceeding — manual entries can always override the default.
- One cart can be associated with multiple primary work orders.
- Required fields (weight or quantity) are pre-configured per product SKU.
- Cart number is entered for each record.

Half-Cart Continuation

> For the **first cart of a session**, the employee must select one of the following — this field is mandatory and cannot be skipped:
>
> - **Standalone cart** — a new, independent production run beginning at this session.
> - **Half-cart (continuing)** — this cart is a physical continuation of a previous session's cart. The system links the current sub-number to the prior cart number, maintaining unbroken physical traceability.

Work Order Close — Restricted (MFA Required)

> Because actual production output is typically less than the theoretical BOM quantity, a work order close function is provided. This action requires special permission and Microsoft Authenticator verification.

Before closing a work order, the authorized user selects one of three options:

Option	Description
No Remainder	All materials are fully consumed. Work order closes cleanly with no leftover raw material.
Scrap Remainder	Remaining raw material is written off. The system auto-distributes the unused quantity evenly across all finished carts for accurate material accounting.
Retain Remainder	Remaining material is preserved for a future run. No adjustment is made to cart records.

Module C Dry Room Operations

Extends existing dry room tracking with production-aligned cart logging.

- **Scan-In:** Records cart number and entry timestamp when a cart enters the dry room. The system automatically retrieves the cart's production record — including any half-cart associations established in Module B — from the barcode; no re-selection is required.
- **Scan-Out:** Records exit timestamp when the cart leaves the dry room.

> Additional dry room workflow details (redry queue integration, time/temperature logging) will be confirmed with the NPIC team during the requirements phase.

Module D Packaging — Material Picking (领料)

Digitizes the handwritten Picking Sheet used to request and receive packaging materials from the warehouse.

Data Captured

- Work ticket number
- Item codes and descriptions (bags, cases, etc.)
- Unit of measure (UOM) and required quantities
- Lot numbers of materials picked
- Picked quantities and worker initials

Output

- Digital pick confirmation replacing the paper Picking Sheet, linked to the packaging work ticket for traceability

Module E Packaging — Shift Production Record (Traveler Form)

Replaces the handwritten Traveler Form (Packing) used to log shift-by-shift packaging output.

Fields Captured (per shift entry)

Field	Description
Date	Production date
Shift	Shift number (1, 2, 3)
Time In / Time Out	Shift start and end times
Bags / Pieces (OWB)	Output quantity in bags or pieces
Cases	Number of finished cases produced
Work Hours	Auto-calculated from Time In / Time Out
Total People	Number of workers on the line
Packing Line	Line identifier (e.g., AB2)
Operator Names	Names of workers on the line for the shift
Labels Printed	Operator initials confirming label printing

Outputs

- Auto-calculated totals: total bags/pieces, total cases, total hours
- Digital supervisor verification (electronic signature)
- Records stored digitally, replacing paper Traveler Form

Module F Packaging — Pallet Tracking & QC

Replaces the per-pallet handwritten form used to record palletizing data and QC inspection results.

Pallet Data

Field	Description
Pallet #	Sequential pallet number within the work order
Stage 2 WO#	Packaging-stage work order number
Date Code	Product date code (Best By / Best Before / MFG / Best Used)
Pieces / Bags	Quantity of product on the pallet
Cases	Number of cases on the pallet
Shift	Shift during which the pallet was packed
Gross Weight Per Pallet	Total weight of pallet + product

Field	Description
Pallet Weight	Tare weight of the empty pallet

QC Records

QC Check	Result Options
Moisture (水份)	Acceptable (Acc) / Rejected (Rej)
Water Activity (水活性)	Acceptable (Acc) / Rejected (Rej)
Metal Detection — FE	Threshold value recorded; Pass / Fail
Metal Detection — NFE	Threshold value recorded; Pass / Fail
Metal Detection — SUS	Threshold value recorded; Pass / Fail
Overall Packaging Inspection Result	Accept (Acc) / Reject (Rej)
Issue Description	Free-text field for non-conforming pallets

Module G Testing & Release

Provides a structured digital workflow for lot testing decisions and product release authorization.

- **Pass / Fail determination** per lot or batch
- **Lot number assignment and tracking** — linked to production and pallet records
- **Redry queue management** — failed lots are automatically flagged and queued for redry, integrating with dry room scheduling
- **Release authorization** — restricted permission requiring digital sign-off before product can move to finished goods

Module H Finished Goods & Warehouse

Manages downstream inventory after packaging is complete and lots are released.

- **FIFO enforcement** — inventory managed first-in-first-out by lot number and date code
- **Case-level tracking** — cases tracked from pallet (Module F) through to outbound
- **Semi-finished goods** — partial production runs tracked separately and linked to open work orders
- **Cross-reference to pallet and lot records** — full audit trail from raw material to finished goods shipment

Module I HR — Attendance & Performance

Because production (Module A/B) and packaging (Module E) already capture employee sign-in/out times, cart output, and shift hours, the HR module draws directly from that data to build attendance and performance records — no duplicate entry required.

Attendance

- Daily attendance automatically derived from sign-in / sign-out records in production and packaging modules
- Shift hours, late arrivals, and early departures surfaced on a per-employee, per-date basis
- Monthly attendance summary report per employee

Performance

- Per-employee output metrics (carts completed, cases packed, hours worked) aggregated from production and packaging records
- Productivity trends over configurable date ranges and by product or work order
- Supervisor can add qualitative notes or evaluations linked to an employee's record

Leave Management

Supervisors can log leave events for their team members. Supported leave types include:

Leave Type	Description
PTO (Paid Time Off)	General paid leave drawn from the employee's accrued balance
Vacation	Pre-planned paid time away
Sick Leave	Unplanned absence due to illness or medical appointment
Personal Day	Short-notice personal time off
Bereavement	Leave following the death of a family member
Jury Duty	Leave for court-mandated jury or witness service
FMLA	Family and Medical Leave Act — extended leave for qualifying medical or family reasons
Unpaid Leave	Approved absence without pay

- Leave entries are reflected in the attendance record, so shift hours are adjusted automatically for the logged date
- HR/Admin can configure accrual rules and view remaining balances per employee

Access

- Supervisors can view attendance, performance, and leave records for their own team, and can create/edit leave entries; HR/Admin access covers all employees
- Export by date range for payroll or review purposes

Module J Document Management & SOP Hub

A centralized, in-system repository for standard operating procedures and operational documents — functioning as an internal shared drive with the additional capability of auto-generating certain SOPs from live production and QC data.

Auto-Generated SOPs

- The system generates draft SOP documents by pulling structured data already in the database — for example, product-specific packaging procedures derived from SKU configurations, or QC checklists assembled from testing module parameters
- Auto-generated documents are versioned and can be reviewed and published by an authorized user

File Upload & Manual Documents

- Supports upload of any file type (PDF, Word, Excel, images, etc.)
- Files are organized by category (e.g., Production SOPs, Packaging SOPs, QC Procedures, HR Policies)
- Full version history: each new upload creates a new version; prior versions remain accessible

Document Lifecycle

- Draft → Review → Published workflow with role-based approval
- Employees can be required to acknowledge reading a document (read confirmation log)
- Expiry / review-due dates can be set per document to prompt scheduled updates

Module K — Optional Sage 100 Integration

> This module is an optional add-on. When enabled, the NPIC system connects directly to the Sage 100 database for real-time data exchange — eliminating manual re-entry of work orders, inventory adjustments, and production completions between the two systems.

Data Pulled from Sage 100

Data	Purpose in NPIC System
Work orders (工单)	Auto-populate work order numbers in production and packaging modules; no manual keying required
Item master / BOM (一级单)	Sync SKU configurations, product descriptions, and raw material codes used throughout the system
Customer POs	Link packaging work orders to customer purchase orders for traceability
Inventory levels	Provide real-time raw material availability when opening a new work order

Data Pushed to Sage 100

Data	Trigger
Production completions (finished goods receipts)	Automatically posted when a work order is closed in Module B
Raw material consumption	Material quantities consumed (from 一级单 reconciliation at work order close) posted back to Sage 100 inventory
Lot numbers	Lot assignments created in Module G (Testing & Release) synced to Sage 100 for end-to-end lot traceability
Pallet / case counts	Final packaged quantities from Module F posted to Sage 100 finished goods inventory

Sync Design

- **Real-time pull:** Work order and item master data is fetched from Sage 100 on demand when a user opens a new record, ensuring the latest data is always used
- **Triggered push:** Production completions and inventory adjustments are posted to Sage 100 at defined workflow milestones (work order close, release authorization) rather than on a polling schedule, minimizing conflicts
- **Error handling:** If a push to Sage 100 fails, the NPIC system queues the record and retries automatically; supervisors receive a notification for any items that require manual resolution

5. Implementation Plan

Phase	Activities	Duration
Requirements Finalization	Confirm workflows, field definitions, and permission matrix with NPIC team; review existing Excel and paper form templates	1 week
Development	Build Modules A—J with role-based access; integrate with Phase 1 authentication and export framework	10—12 weeks
Internal Testing	End-to-end system testing using NPIC production data and workflows	2 weeks
Parallel Run	Run new digital system alongside existing Excel and paper forms; compare outputs and resolve discrepancies	2 weeks
Training		1 week

Phase	Activities	Duration
	Train supervisors and employees on each module; document SOPs for each role	
Go-Live	Full cutover to digital system; retire Excel spreadsheet and paper forms	—

6. Deliverables

- Phase 2 modules (A–J) integrated into the existing NPIC operations platform
- Role-based access control: Employee / Supervisor / Manager / HR / Admin
- Microsoft Authenticator MFA gating for restricted operations (work order close, release authorization)
- Shift summaries and per-employee productivity and attendance reports
- Pallet and QC records with digital audit trail
- Data export by date range or product (leveraging Phase 1 export framework)
- FIFO finished goods inventory management
- Document management hub with auto-generated SOPs and version-controlled file library
- (Optional) Module K — Sage 100 bi-directional integration: real-time work order pull and automated production completion / inventory push
- Training documentation for each module and role

7. Expected Benefits

Benefit	Impact
Eliminate manual Excel production logging	Reduced data entry errors; real-time visibility into output per shift and per worker
Replace all three paper packaging forms	Digital records, searchable and audit-ready; supervisor signatures captured electronically
Automated productivity metrics	No manual calculation; instant shift summaries with individual output data
End-to-end lot traceability	From raw material (一级单) through drying, packaging, and palletizing — all linked in one system
Digital QC records per pallet	Instant access to moisture, water activity, and metal detection results; no paper filing required
FIFO inventory enforcement	Reduces material waste and risk of shipping out-of-date product

Benefit	Impact
Zero–effort HR attendance tracking	Attendance and performance data auto–derived from production records; no separate time–clock system needed
Centralized document control	Auto–generated SOPs stay in sync with system configuration; all documents versioned, searchable, and acknowledgement–tracked
Seamless integration with Phase 1	Single platform for all QC inspection and production operations; no duplicate data entry